

Alejandro Morera Alvarez

AI LEAD AT PROFUTURO · ML FORESIGHT EXPERT AT NOVALAN

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EDUCATION

PhD Researcher in Mathematics January 2027

Technische Universität München

Project: *Learning and Dynamics in Stochastic Multi-Agent Systems* — joint project of the groups of Prof. Martin Bichler, Prof. Christoph Knochenhauer, Prof. Christian Kuehn, and Prof. Jingui Xie.

Double M. Sc. in Mathematics and Applied Mathematics April 2024 – August 2026

École Polytechnique Fédérale de Lausanne & Technische Universität München

German GPA 1.1 – “Passed with High Distinction” Swiss GPA 5.65 – “Outstanding”

Highlights: Stochastic Analysis, Fractal Geometry, (Random) Dynamical Systems, Optimal Transport.

Thesis: *Mean-Field Limits for Stochastic Interacting Particle Systems on Digraphs Measures: Convergence, Regularity and Applications to Machine Learning Dynamics on Large Networks*

B. Sc. Mathematics October 2022 – March 2024

Technische Universität München · German GPA 1.3

Thesis: *Borel Functional Calculus and the Spectral Theorem*

B. Sc. Engineering Science October 2018 – June 2023

Technische Universität München · German GPA 1.709

Thesis: *Borel Functional Calculus and the Spectral Theorem*

American International School Salzburg — AP Scholar with Distinction American GPA 4.031/4.3

Colegio Vista Hermosa — Duc in Altum Award (3 times) GPA 97.5/100

WORK EXPERIENCE

AI Lead — Profuturo May 2026 – Now

Lead a team in strategic AI developments from the technical side and provide a bridge for communication between the team and several company areas, including legal, financial, risk, and investment.

ML/AI Foresight Expert — Novalan April 2026 – Now

Use ML/AI models and tools to predict desired phenomena; implement AI automations to build systems within the company which simplify analyses and processes.

AI Analyst — Profuturo August 2025 – May 2026

LLM, RAG and Automation Expert in the AI Team. Built RAG pipelines for AI agents with distinct tools, strategies, and architectures.

Teaching Assistant — EPFL September 2024 – June 2025

Modules: Probabilité (B. Sc. Mathématiques), Analysis III (B. Sc. Génie mécanique), Analyse IV (B. Sc. Physique).

Teaching Assistant — TUM October 2020 – August 2024

Modules: Probability Theory (M. Sc. Mathematics), Analysis 3 (B. Sc. Mathematics), Abstract Algebra (B. Sc. Mathematics), Analysis 1 (B. Sc. Physics), Real Analysis and Linear Algebra (B. Sc. Engineering Science).

RESEARCH EXPERIENCE

MASTER THESIS	February 2026 – August 2026
Mean-Field Limits for Stochastic Interacting Particle Systems on Digraphs Measures: Convergence, Regularity and Applications to Machine Learning Dynamics on Large Networks	
PROJET DE MASTER	February 2025 – June 2025
Fyodorov–Bouchaud Formula and Beyond	
BACHELOR THESIS	April 2023 – June 2023
Borel Functional Calculus and the Spectral Theorem for Bounded Operators	
RESEARCH PROJECT	October 2023 – December 2023
Convergence to Equilibrium: Entropy Methods in PDEs and Stochastic Analysis	
R PROGRAMMING + RESEARCH PROJECT	September 2023 – February 2024
Education in Different Countries	

SEMINARS & WORKSHOPS

Presented talks on Elliptic Curves, Infinite-dimensional Vector Spaces, Selected Topics in Functional Analysis, Applied Asymptotic Analysis.

Attended Particle Systems in Random Environments (Summer School, 2024); research talks at EPFL (2024–2025) and ITAM (2025–2026).

LANGUAGES

Spanish – Native	English – C2	German – C1	French – B1-B2	Norwegian – A2
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COMPUTER & AI SKILLS

GENERAL	Autodesk AutoCAD Professional, Autodesk Inventor Professional (Computer-Aided Modelling of Products and Processes I), CATIA, Microsoft Office, MATLAB, Julia, and R.
AI & AGENTS	Claude Code- and AI-native development; Copilot Studio, Power Automate, n8n, Microsoft Azure AI, OpenAI AgentKit and Agents SDK, LangChain, LangGraph, LlamaIndex, tool/function calling, and MCP client/server integration.
HYBRID RAG	Semantic vector search plus BM25/keyword retrieval; embeddings, document chunking, metadata filtering, query rewriting, reranking, context assembly, grounded citations, and RAG evaluation.
DATA & APIS	SQL, PostgreSQL/pgvector, Qdrant, and Azure AI Search; FastAPI, Pydantic, REST/JSON APIs, OpenAI/Azure OpenAI APIs, Microsoft Graph/SharePoint integrations, OAuth 2.0, and API-key authentication.
ENGINEERING	Python, Pandas, document parsing/OCR, Git, Docker, automated testing, logging/observability, and secrets management.

OTHER EXPERIENCE

Grupo ADO Summer Course	Mexico City, Mexico · June – July 2017
Understanding of how the company works and how smaller subsections interact with each other to make the group as a whole work.	
Student Tutoring and Funding Administration	May – July 2017
Administered funding for a student tutoring initiative, directly facilitating educational support for approximately 170 students across mathematics, physics, and chemistry, improving students’ pass rate.	
Raised funds to build a well in a water-lacking community.	